

□ Bird Species Observation Analysis Report

Forest vs Grassland Habitat Study

11 National Park Admin Units | 17,077 Observations | 2018–2019

Executive Summary

This report presents a comprehensive analysis of bird species observations collected from Forest and Grassland habitats across 11 National Park administrative units. The dataset contains a total of 17,077 bird observation records spanning the years 2018 and 2019. Both habitats show nearly identical species richness, with the Forest recording 108 unique species and Grassland recording 107 unique species. The study reveals critical conservation concerns, with 8 at-risk species identified on the PIF Watchlist — most notably the Wood Thrush with 309 recorded observations.

1. Dataset Overview

The dataset was collected across two distinct habitat types — Forest and Grassland — within 11 National Park administrative units located in the northeastern United States.

Metric	Value
Total Observations	17,077
Forest Observations	8,546
Grassland Observations	8,531
Unique Species (Forest)	108
Unique Species (Grassland)	107
Admin Units Covered	11
Years of Data	2018 – 2019

Metric	Value
At-Risk Species (PIF Watchlist)	8

2. Species Diversity: Forest vs Grassland

Finding: Forest has slightly higher species diversity

The Forest habitat recorded 108 unique bird species, compared to 107 in the Grassland — a difference of just 1 species. This indicates that both habitats are nearly equal in terms of species richness, suggesting that both ecosystems provide important and comparable ecological niches for bird populations.

Habitat	Unique Species	Total Observations
Forest	108	8,546
Grassland	107	8,531

Key observation: 88 species were found in BOTH habitats, while 20 species were exclusive to Forest and 19 species were exclusive to Grassland. This overlap demonstrates that many bird species are habitat-generalists, while a significant number have specific habitat preferences.

3. Bird Activity by Admin Unit (Park)

Finding: ANTI has the highest bird activity

Among all 11 administrative units, ANTI (Antietam National Battlefield) recorded the highest number of bird observations with 3,921 sightings — accounting for nearly 23% of all observations. MONO ranked second with 3,385 observations, followed by PRWI with 2,463 observations.

Admin Unit	Total Observations	% of Total	Rank
ANTI	3,921	22.9%	1st
MONO	3,385	19.8%	2nd
PRWI	2,463	14.4%	3rd
MANA	2,276	13.3%	4th
CHOH	2,202	12.9%	5th
CATO	805	4.7%	6th
NACE	684	4.0%	7th

Admin Unit	Total Observations	% of Total	Rank
HAFE	539	3.2%	8th
GWMP	386	2.3%	9th
ROCR	289	1.7%	10th
WOTR	127	0.7%	11th

4. Seasonal Bird Activity

Finding: Summer is the peak season for bird sightings

Summer dominates bird activity with 11,481 observations — representing 67.2% of all recorded sightings. Spring follows with 5,596 observations (32.8%). Interestingly, no observations were recorded during Autumn or Winter, suggesting that data collection was limited to the warmer breeding and migratory seasons.

Season	Observations	% of Total
Summer (Jun–Aug)	11,481	67.2%
Spring (Mar–May)	5,596	32.8%
Autumn (Sep–Nov)	0	0%
Winter (Dec–Feb)	0	0%

Recommendation: Future data collection should be extended to Autumn and Winter months to capture year-round bird activity patterns, including migratory species passing through the region.

5. Conservation: At-Risk Species (PIF Watchlist)

Finding: 8 at-risk species recorded — Wood Thrush most frequent

A total of 378 observations (2.2% of all records) belong to species listed on the Partners in Flight (PIF) Watchlist, indicating species of conservation concern. Eight unique at-risk species were identified across both habitats, with the Wood Thrush being the most frequently observed at-risk species with 309 sightings.

Species (Common Name)	Observations	Conservation Status
Wood Thrush	309	PIF Watchlist
Worm-eating Warbler	31	PIF Watchlist

Species (Common Name)	Observations	Conservation Status
Prairie Warbler	25	PIF Watchlist
Cerulean Warbler	7	PIF Watchlist
Kentucky Warbler	2	PIF Watchlist
Willow Flycatcher	2	PIF Watchlist
Blue-winged Warbler	1	PIF Watchlist
Red-headed Woodpecker	1	PIF Watchlist

The Wood Thrush alone accounts for 81.7% of all at-risk species observations, making it a priority species for conservation monitoring in these national park units.

6. Environmental Conditions & Bird Activity

Finding: Moderate temperatures (21–27°C) yield the most observations

Temperature ranged from 11°C to 37.3°C across all observation sessions, while humidity ranged from 7.3% to 98.8%. Analysis of temperature bins shows that the highest bird activity occurred at moderate temperatures between 21.5°C and 26.8°C, with 7,401 observations — suggesting birds are most active during mild, comfortable conditions.

Temperature Range	Observations	Activity Level
11.0°C – 16.3°C	1,154	Low
16.3°C – 21.5°C	5,888	High
21.5°C – 26.8°C	7,401	Peak
26.8°C – 32.0°C	2,216	Moderate
32.0°C – 37.3°C	418	Low

This pattern suggests that extreme heat (above 32°C) significantly reduces observable bird activity, likely due to birds seeking shade and remaining less vocal during the hottest parts of the day.

7. Bird Identification Methods

Finding: Singing is the most common identification method

Among the three identification methods recorded, Singing was the most frequently used with 9,863 observations (57.7%), followed by Calling with 4,061 observations (23.8%), and

Visualization with 3,151 observations (18.4%). This indicates that most birds in these habitats are detected by their vocalizations rather than direct visual sighting.

Identification Method	Observations	% of Total
Singing	9,863	57.7%
Calling	4,061	23.8%
Visualization	3,151	18.4%

The dominance of singing as an identification method highlights the importance of acoustic monitoring in forest and grassland bird surveys. It also suggests that passive acoustic recording devices could significantly enhance future data collection efforts.

8. Habitat-Exclusive Species

Finding: 20 Forest-only and 19 Grassland-only species identified

While 88 species were found in both habitats, a notable number of species showed strict habitat preference. The Forest habitat had 20 exclusive species, while the Grassland had 19 exclusive species — demonstrating that both ecosystems support unique ecological communities.

Top Forest-Exclusive Species

Species	Observations
American Goldfinch	147
Hooded Warbler	64
Louisiana Waterthrush	43
Worm-eating Warbler	31
Black-throated Blue Warbler	9
Cerulean Warbler	7
Double-crested Cormorant	7
Osprey	7

Top Grassland-Exclusive Species

Species	Observations
American Goldfinch	362

Species	Observations
Bobolink	103
Tree Swallow	73
Rock Dove	28
Unidentified Swallow	22
Blue Grosbeak	15
American Kestrel	8
Purple Martin	6

9. Conclusions & Recommendations

Based on the analysis of 17,077 bird observations across Forest and Grassland habitats in 11 national park units, the following key conclusions are drawn:

- Both Forest and Grassland habitats support nearly equal species diversity (~108 species each), confirming their ecological importance.
- ANTI (Antietam National Battlefield) is the most critical monitoring site with 3,921 observations and should receive priority conservation attention.
- Summer is the peak season for bird activity — surveys should be intensified during June–August for maximum data yield.
- The Wood Thrush is the most at-risk species and requires dedicated monitoring and habitat protection efforts within these parks.
- Moderate temperatures (21–27°C) coincide with peak bird activity, which should be considered when scheduling future surveys.
- Singing is the dominant detection method — acoustic monitoring equipment should be deployed to supplement human observers.
- Habitat-exclusive species confirm that both Forest and Grassland ecosystems must be preserved independently, as each supports unique biodiversity.

End of Report

Data Source: Bird Monitoring Program — National Capital Region Network (NCRN)